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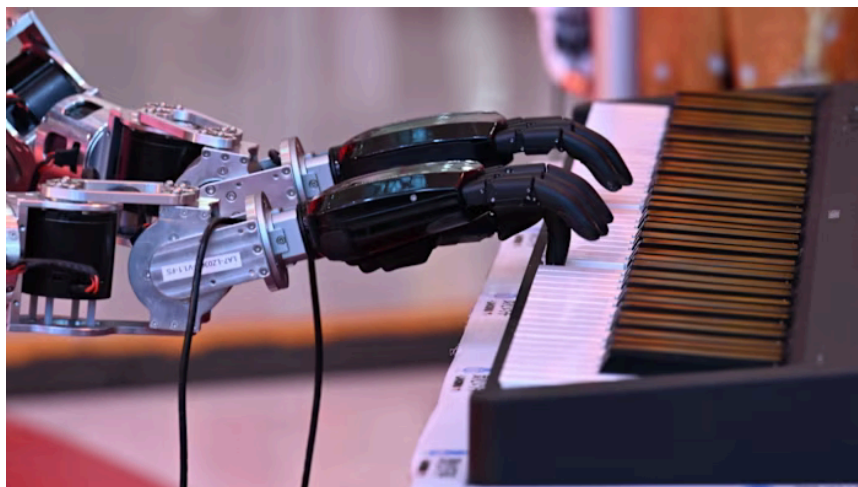
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FINANCIAL TIMES

Opinion **Unhedged**

The Citrini fuss exposes a market looking for an excuse to fall

Fear > insight

ROBERT ARMSTRONG[Add to myFT](#)

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Robert Armstrong

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44

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AI-doomer fan fic FTW

Yesterday, following the recent pattern, the shares of various companies that might be vulnerable to AI disintermediation fell — software, asset management and, joining the party for the first time, banks. The most popular explanation for the renewed jitters was a [blog post](#) from Citrini Research about how AI might get a lot of high earners fired and tank the economy. From [the FT](#):

Investors have recently seized on social media rumours and incremental developments by small AI companies to justify further selling, with a widely circulated blog post by Citrini Research over the weekend describing how AI could hypothetically push the US unemployment rate above 10 per cent by 2028, proving the latest catalyst.

The most important thing about the post is not what it says. It is that the stock market has reached the point where blog posts cause significant [stock moves](#), or at least where people think that they do. It is often true that markets sell off for reasons that are obscure or all but unknowable, and people like me have to feel around for an explanation. This may be one of those cases. Either way, the Citrini fuss is further evidence that we are in an expensive market that is looking for an excuse to fall, for reasons that are probably wider than just AI.

But what about the post caught the market's attention? It takes the perspective of several years in the future, from which it reflects on a recession and financial crisis that have already happened. "What follows is a scenario, not a prediction . . . The sole intent of this piece is modeling a scenario that's been relatively underexplored," it begins. But the opposite is true. The post took off because it is an extreme version of one of the most explored scenarios of all time: robots turning on their creators. It's been explored to death at least since Karel Čapek introduced the word robot and, at the very same moment, predicted that robots might

cause a lot of trouble. That was over a hundred years ago. More than a century before that, some weavers in Nottingham, under the banner of the fictional General Ned Ludd, explored the theme by smashing up some textile mills.

On to the post's argument. The central scenario is that AI destroys jobs and income, but does not replace them, dragging the economy into recession and markets into a crisis:

The owners of compute saw their wealth explode as labor costs vanished. Meanwhile, real wage growth collapsed . . . white-collar workers lost jobs to machines and were forced into lower-paying roles . . . When cracks began appearing in the consumer economy, economic pundits popularized the phrase “Ghost GDP”: output that shows up in the national accounts but never circulates through the real economy . . . The velocity of money flatlined. The human-centric consumer economy, 70 per cent of GDP at the time, withered . . . AI capabilities improved, companies needed fewer workers, white collar layoffs increased, displaced workers spent less, margin pressure pushed firms to invest more in AI, AI capabilities improved . . . It was a negative feedback loop with no natural brake . . .

The irony of this was that the AI infrastructure complex kept performing even as the economy it was disrupting began deteriorating. NVDA was still posting record revenues. TSM was still running at 95 per cent+ utilization. The hyperscalers were still spending \$150-200 billion per quarter on data center capex. Economies that were purely convex to this trend, like Taiwan and Korea, outperformed massively.

Something struck me as odd about this picture, although (with my non-existent formal education in economics) I struggled to articulate what it was. It feels like financial commentary that considers only one side of a trade (“stocks went down as sellers outnumbered buyers”, etc). Citrini

pictures a world of massively increasing productivity accompanied by a collapse in consumption. Does that make sense?

Joseph Steinberg, an economist at the University of Toronto, helped me flesh out my intuition that something is amiss.

“The first part of the argument I hope my economics students would flag is the bit about ‘ghost GDP’,” he told me. What does it mean for output to “show up in the national accounts but never circulate through in the real economy”? If GDP is rising — all those robots out there making stuff, faster and faster — then something on the other side of the national account identity has to be rising, too. The possibilities are consumption, investment, government spending or net exports. In the Citrini scenario, consumption is falling, fast. So is government spending rising (on the basis of taxing or borrowing from who, exactly)? Or exports — to other countries undergoing the same crisis?

None of that makes loads of sense, so that leaves investment. But investment, Steinberg points out, is only sustainable on the basis of future consumption; “it only makes sense to invest in AI if there is income to buy these things the AI is generating”.

It seems to me there has to be a distributional element to the argument that Citrini does not fully flesh out. Income and consumption rises, but this happens somewhere that doesn’t benefit your average US white-collar worker. Maybe it is all in the “economies that were purely convex to this trend, like Taiwan and Korea”. Or maybe it all goes to a few, very annoying gagillionaires in Silicon Valley. In this scenario, we all end up bringing very, very expensive lattes to Peter Thiel. Or perhaps Thiel’s wealth is seized by the government and redistributed. But it’s hard to see how the Thiels of the world stay rich for long in the world Citrini envisions, because their AI investments stop paying off when consumption collapses.

None of this is to say that I believe that the AI revolution, like past technological regime changes, must in the end create more jobs and making all of us richer. I don’t. What I am

claiming is that the Citrini account of the risks seems incoherent (perhaps readers with a better grasp of macroeconomics can make better sense of it?). The post went viral not because it elucidates the present situation, but because it speaks to ancient fears.

One good read

“A fire chief may have [the authority](#) to close roads during an emergency but that doesn’t imply that the fire chief has the authority to impose road tolls.”

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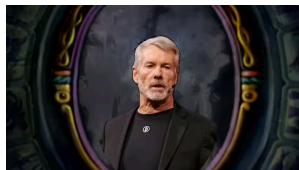
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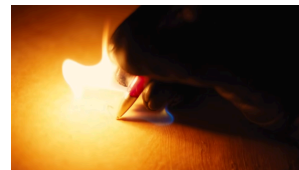
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Newest

— **Merrion** 5 MINUTES AGO

The impact of AI on employment will be to further spread the K shaped economy such that those in the top leg on an income basis will be pushed to the lower leg.

It makes sense that this will reduce the consumption capacity of the economy as a whole. Whether this leads to a runaway system and permanent recession remains to be seen but given that the economy is a non linear system it is very difficult to model or predict.

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— **bambos** 17 MINUTES AGO

I think the US will be very lucky if the unemployment rate is "only" 10% by 2028.

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— **The Missing Chord** 42 MINUTES AGO

What a brilliant, thoughtful piece.

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— **Bassam** 59 MINUTES AGO

Mark Meldrum raised the notion that while unemployment may rise, wages will increase as well for experts and lead to a situation not unlike the 1950s where a one-wage-household is sustainable. hence

the lower participation rate.

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+ **manticore** 50 MINUTES AGO

— **Stevie1** 1 HOUR AGO

I had been hoping you would explain why the market is looking for an excuse to fall...

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— **Don Qazaq** 1 HOUR AGO

"...it speaks to ancient fears."

This encapsulates the current prevailing sentiment: fear.

But not everywhere: as US and European investors tremble in fear, Taiwanese and Korean investors plow ahead.

Do they know something we don't? No. They just refuse to be driven by fear.

Europeans fear Trump and a bursting AI bubble.

They fear a loss of service jobs.

Trump is irrelevant. He is given relevance only because we allow it.

Just like any bully, he is a coward who backs down when challenged and fades when ignored.

AI isn't going anywhere. Our economies will evolve to take advantage of the new powerful efficiencies it offers. No bubble will burst, but there will likely be some consolidation.

Just like in any industrial or technological revolution, people will have to adjust.

The sooner we stop shaking in our boots, the better off we'll be.

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— **Zambean** 1 HOUR AGO (Edited)

When I tried to reassure my teenage son that AI fears are overblown because when they first invented the car, they worried that all the people making carriages and buggy whips would lose their jobs , but it all worked out okay.

He wasn't buying it.

'No dad, you don't get it - you're the horse!'

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+ **Auld Scots Engineer** 27 MINUTES AGO

— **TheRealDebate** 1 HOUR AGO

I'm about as far from being a socialist as it gets, but it seems obvious to me that AI agents should be taxed and help fund human pensions.

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+ **Adam Pharaoh** 56 MINUTES AGO

— **Kplk** 2 HOURS AGO

King GeorgeV once asked a postman what he did, 'I knock on doors and push letters through the door, your majesty'. "Oh dear", said the King, "It wouldn't do, if everyone did that, would it?"

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— **KickoCameron** 2 HOURS AGO

Ex bankers could gain employment in the tariff prediction market. No AI, no matter how advanced, would fare better than a human in this impossible task.

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— **Just a thought** 2 HOURS AGO

Could it be that the AI algos read the report and believed it to be true?

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— **Jenkins** 2 HOURS AGO

...white-collar workers lost jobs to machines and were forced into lower-paying roles...

But what if - scary thought I know - some of those low paying roles became slightly higher paying roles? After all some of them (teachers, carers, healthworkers) are what we will need most of as we age and need the declining number of young to be able to step up in the new economy. Others like service jobs mean more expensive lattes, sure, but do we really think subsistence wages are the only equilibrium in a modern economy?

Of course, making that happen means a big political shift. Personally I think it means wealth taxes in some form. And I don't think that reducing the amount of GDP that is accounted for by the kind of white collar jobs which effectively add transaction costs to the economy (as any fule kno, these are deadweight losses to the wider economy) is a bad thing.

— **Hughby1** 2 HOURS AGO

‘In an expensive market. .’

I remember, I think, the anecdote of Baruch selling out of the stock market because elevator operators or shoeshines were offering stock tips

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— **Nixer65** 2 HOURS AGO

There are two choices -

1. AI is as successful as is being priced in. In which case the economy collapses.
2. AI creates an incremental improvement in productivity, in which case we likely see a similar path following the dot com pattern.

Why?

If AI replaces people en-mass where will the money come from to buy your products? Given the endemic cultural view in the US that any social funding is inherently evil I can't see Universal Basic Income being rolled out anytime soon. Hence, economic collapse.

Or alternatively we follow the "hype cycle" and the AI bubble takes the stock market out and this becomes a repeat of any other tech boom and bust.

I would suggest path 2 is preferable.

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+ **tiktaalik descendant** 1 HOUR AGO

+ **Satisficing** 1 HOUR AGO

— **Uler** 2 HOURS AGO

Entirely possible that we end up in a world of lower consumption for most, and billionaires having more (or larger) yachts (or spacecraft or planes or whatever).

But you have to wonder how likely it is. Even relatively small distributional shifts in the current world are sufficient for national protests. Larger ones (including those caused by war/famine) nearly

always result in popular uprisings, regime change, or civil war.

I don't see a clear way for the billionaire oligarchy to emerge. Maybe if musk can get his million humanoid robots working, but 1m is still far fewer than 8bn plus people...

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+ manticore 57 MINUTES AGO

— Benjamin 2 HOURS AGO

The second long quote reminds me of Marx - explaining that as wealth accrues to fewer and fewer capitalists, previous capitalists are forced into the proletariat. Not saying I agree, just saying that this argument has surfaced before!

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— JM2845 3 HOURS AGO (Edited)

Think most of the critiques of the Citrini piece take it too literally and in doing so miss the point. The prevailing consensus view on AI has been that, similar to the advent of the Internet and PCs, a new class of jobs will be created that human beings can pivot to, but it's worth asking - what if that isn't the case with AI? We may be at the point in the cycle where the market needs to see evidence of this pivot before resuming the AI bull thesis, especially because the harms of AI (disruption, displacement, job loss) are becoming more apparent.

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— neil tritton 3 HOURS AGO

Excellent piece

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— Zola.22 3 HOURS AGO (Edited)

AI and AI-devices are tools. If they prove to be as useful and revolutionary as their proponents claim and their detractors fear, then they will soon supply substantially all the products and services that we might require or wish to have.

If so, we will find ourselves largely relieved of the burden of making the products and services that we require or wish to have. The problem of economic scarcity will be largely solved: AI and AI-devices will make and do everything for us.

Rather than induce a depression, an AI-revolution of this kind would herald unprecedented increases in productivity and output.

At the same time, such a radical transformation might well occasion *ever greater inequality of wealth and income*.

Even if we somehow solve that problem, our liberation from work might prove profoundly unsettling and disorienting to many: we humans are hard-wired to work to sustain ourselves and provide for our loved ones. Many of us find meaning in our work and might languish after our services are no longer required or requested.

Regardless, no one will put this genie back into the bottle. So we better start to ponder and address these kinds of issues. Clamoring to protect certain professions or trades from AI is as hapless and futile as it was when makers of horse-buggies sought protection from the automobile.

As for me, I am skeptical that AI will *replace* human labor. Rather, humans will use it to improve their output by leaps and bounds: we will use it, or our descendants will, to make discoveries and inventions; improve human understanding in every field and in fields that we cannot yet anticipate; and make better products and services.

I worry only about bad actors using AI for sinister purposes. Never underestimate the perfidy, greed, and malice of the worst among us.

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 + Rushwarp 3 HOURS AGO

— Fouquet 3 HOURS AGO

Sorry Robert I am not with you on this one.
Firstly
Taxes/ legal rights play a big role in distorting the playing field.
Replacing labour with AI is tax deductible
Hiring labour instead accumulates tax liabilities tax collection on behalf of the government plus holiday pay, sickness pay, maternity leave, pension rights etc
As long as this imbalance exists I foresee growing unrest.

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— JohannesIgnatius 4 HOURS AGO (Edited)

Last week it was Matt Shumer. 48 hours ago it was Citrini. Last night IBM apparently fell 10% on reports that Claude can do Cobol. Things are getting increasingly unhinged, and at speed.

are getting increasingly unminged, and at speed.

As yet, the only evidence for the revolutionary impact of 'AI' is excited anecdotes, and the conclusions of attempts so far to validate those anecdotes with systematic research have been at best mixed.

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— **Please settle all grudges** 4 HOURS AGO

Thank god, an economist trying to form a sane view of AI ...

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— **Hfjjggf** 4 HOURS AGO

Aren't we also forgetting the ageing demographics of the Western world and the fact that we actually need to find massive productivity improvements or we're in big big trouble. AI displaced labour will likely be absorbed by healthcare given demand will only trend higher as boomers require ever more of it to keep aged bodies functioning. We are seeing this in the stats already. Healthcare has been the only area of employment growth in the US since start of 2024. Suspect that trend to continue...

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— **DontpanicMrMainwaring** 4 HOURS AGO

It seems to me the report's error is to assume that the AI companies can continue to invest if the wider economy tanks. The companies they sell AI tools to will have no employees or paying customers so they will stop buying AI tools or maybe trading.

There will be no ghost GDP. Facebook and the others rely on the wider economy for their cash piles.

A more likely scenario in view is that investors and lenders start to realise that the payback on all that tied up money is much longer than expected leading to crash in company valuations.

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— **John_CEE** 4 HOURS AGO (Edited)

Considering AI impact on jobs and overall society, we shouldn't focus just on GDP, investments, etc., but also including social and political dimensions. In the 19th and beginning of the 20th century, we had a very similar trend of monopolies in rail (as well as other sectors of the new economy) where new CAPEX-intensive technology triggered not only rapid monopolization and growth but also strong political backlash.

If we think that AI will be such a potent force impacting society, jobs, etc., we should add political risk to our models. Trump is one of such „political factors” (similar to other presidents from the past, particularly the 19th/early 20th century) as he „attack” several old-fashioned business orthodoxies of the Republican Party. So if someone is forecasting super good returns from AI but also a major crisis on the job market, the „political” drag on all those returns should be included.

And this is probably why people like Thiel, Musk are hedging its position with good relations with radicals like JD Vance. They try to „manage” the backlash risks in advance (like „cultural wars” as a substitution for real discussion about the economy) but still I wonder why they believe - we are back to the GDP & consumption and the article above - that the economy as a whole will be able to sustain never-ending growth of „Musk and co” monopolies without strong white-collar jobs or radical change in taxation/redistribution.

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+ Dr.Rationality 2 HOURS AGO (Edited)

— Mr. D 5 HOURS AGO

"Citrini pictures a world of massively increasing productivity accompanied by a collapse in consumption. Does that make sense?:

The issue here that there is a difference between real and nominal growth. The collapse in consumption is nominal while the growth is real.

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— Darth Blair 5 HOURS AGO

"it only makes sense to invest in AI if there is income to buy these things the AI is generating"

Yes, Professor, read that back to yourself slowly a couple of times. Maybe perhaps over a cup of tulip tea.

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— No.11 5 HOURS AGO

Excellent piece, that old adage of "never let a few facts get in the way of a good story."

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— JackSawyer 5 HOURS AGO

Which also means, AI is likely to already have collapsed along the way as you mentioned that a counterpart to the equation has to happen as well. Either ways, seems like upside to AI is extremely capped unless it finds a way to add value alongside humans rather than replace them. Do badly a job(extremely likely given the unsustainable economic equation for AI at the moment and grossly exxagerated capabilities Anthropic is aggressively pumping at the moment) and the AI bubble pops. Do too well a job that it replaces humans and we start a negative feedback cycle that collapses economies. If humans are really replaced permanently as AI advances, the pain will immediately showup at the very same companies that replace them with a more than proportionate hit to their revenues due to the multiplier effect.

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— Impatient taxpayer 5 HOURS AGO

It's pretty obvious - do keep up. Investment keeps rising as robots set up factories to produce more robots and spaceships. All part of a cunning plan to colonize Mars (for starters).

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— lllllun 5 HOURS AGO

I agree

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